FRANCISCO MORENO DE LA CRUZ

Backend Developer

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Summary

Backend Developer and Higher Technician in Multiplatform Application Development, with knowledge of C, C++, Python, SQL and Linux. Additional hands-on training at 42 Madrid – Fundación Telefónica, with projects in systems programming, networking and algorithms. Keen to further develop my career in backend development and software architecture.

Professional Experience

Backend Developer - Mirelia Digital Talent

03/2024 – 08/2024

TypeScript • Prisma ORM • PostgreSQL • Docker • Git

Madrid, Spain

- Developed the backend using TypeScript, Prisma ORM and PostgreSQL.
- Designed and implemented REST APIs, endpoints, business logic and data access.
- Designed database models and relationships using Prisma ORM.
- Configured the environment with Docker and managed code and version control using Git.

Education

Software Developer

11/2024 – 06/2026

42 Madrid – Fundación Telefónica

- C, C++, Linux, Python, Shell, SQL, Docker, Git, TCP/IP, Algorithms & Data Structures
  - Developed an IRC chat server in C++, implementing client-server communication via sockets and managing multiple connections, channels and commands.
  - Developed a GPS application in Python, working with data processing, routing logic and geographic information management.
  - Implemented a Minesweeper-solving algorithm in Python using board analysis, logical inference and probability calculations.

Higher Vocational Diploma in Multiplatform Application Development

09/2022 – 07/2024

CampusFP

- Python, SQL, Java, PHP, Git, Algorithms & Data Structures
  - Developed an Excel-style administrative application in Python, connected to an SQL database to query edit and manage information in a structured way.

Computer Engineering

09/2019 – Present

University of Alcalá de Henares

- C, C++, SQL, Python, Algorithms & Data Structures, Operating Systems, Statistics

Higher Vocational Diploma in Industrial Automation and Robotics

09/2017 – 07/2019

IES Luis de Lucena

Skills

Programming Languages: C • C++ • Python • SQL • Shell

Systems & DevOps: Linux • Git • Docker

Software Engineering: Algorithms & Data Structures • Object-Oriented Programming • Software Architecture • Networking

Soft Skills: Self-directed Learning • Problem-solving • Teamwork • Critical Thinking

Certificates

Python Associate • Link	11/2026 – 11/2028
SQL Associate • Link	11/2026 – 11/2028
MongoDB RAG • Link	09/2026
MongoDB AI • Link	09/2026

Selected Projects

Minishell – C

LinkedIn | GitHub

Developed a Bash-inspired shell supporting command parsing, environment variables, and built-in commands. Implemented pipes, redirections, process management, signals, and exit codes.

Minesweeper AI – Python

LinkedIn | GitHub | PyAutoGui

Developed an algorithm to solve Minesweeper games through pixel analysis, logical inference, and probability calculations. Implemented board analysis, decision-making, and safe-move selection.

Languages

Spanish	Native
English	B2